Chapter 1: Introduction to Programming & C

1. Programming Languages

- **Definition:** A formal set of instructions allowing humans to communicate with computers.
- **High-Level Language (HLL):** Human-friendly, uses English-like syntax, and is **portable** (e.g., C, Python).
- **Low-Level Language (LLL):** Machine-friendly, hardware-dependent. Includes **Machine Code** (Binary) and **Assembly** (requires an Assembler).
- **The C Identity:** C is a **High-Level Language**, but often called **Middle-Level** because it can manipulate hardware directly while remaining portable.

2. Software Categories

- **System Software:** Manages hardware and provides a platform (e.g., Operating Systems, Compilers, Device Drivers).
- **Application Software:** Performs specific user tasks (e.g., Chrome, VS Code, Word).
- *Note:* The **C Compiler** is System Software. **VS Code** is Application Software (an IDE).

3. Computer Architecture (Block Diagram)

- **CPU (Central Processing Unit):** The "brain" consisting of:
  - **ALU (Arithmetic & Logic Unit):** Performs calculations and logic.
  - **CU (Control Unit):** The manager; fetches, decodes, and sends control signals.
  - **Registers:** Internal, high-speed storage inside the CPU.
- **Memory Unit (RAM):** Separate from the CPU; stores data/instructions currently in use.

4. Key Features of C

- **Portability:** Source code can be compiled on different platforms (Windows/Linux/Mac) without rewriting logic.
- **Structured:** Uses control statements (`if`, `while`) to keep logic clear and organized.
- **Modular:** Programs are broken into smaller, reusable **functions**.
- **Efficient:** Provides fast execution due to its "middle-level" nature.

5. Planning Tools

- **Algorithm:** Step-by-step logical procedure in plain English.
- **Flowchart:** Visual representation of an algorithm using standard shapes:
  - **Oval:** Start/Stop (Terminal).
  - **Parallelogram:** Input/Output.
  - **Rectangle:** Processing (Math/Calculations).
  - **Diamond:** Decision (Must have Yes/No branches).

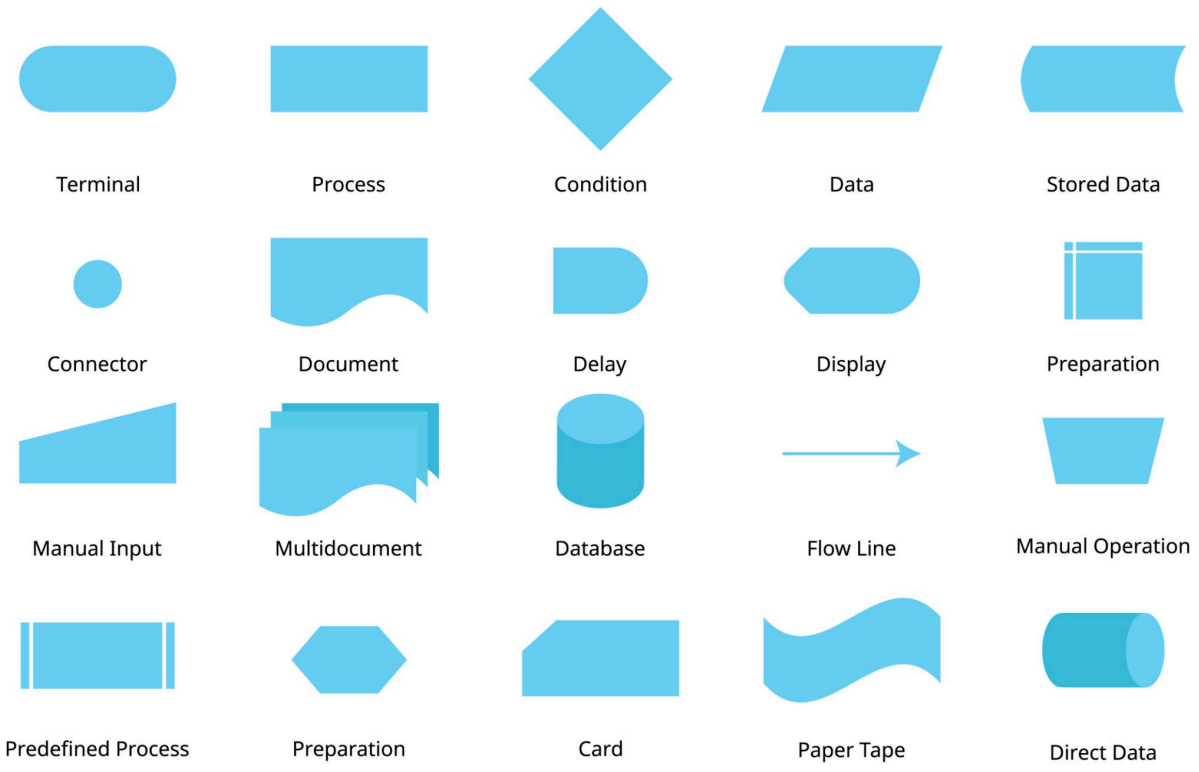

fig: basic flowchart symbol with meanings to create the flowchart